Object detection in natural environment

by Ethnic-Local-Pace

Problem statement

Object detection in harsh weather conditions

How do we cover the problem?

The Barrier

Harsh environmental conditions creating “real world blindness” for vehicles

The Solution

Denoising into clear images for higher accuracy

The Strategy

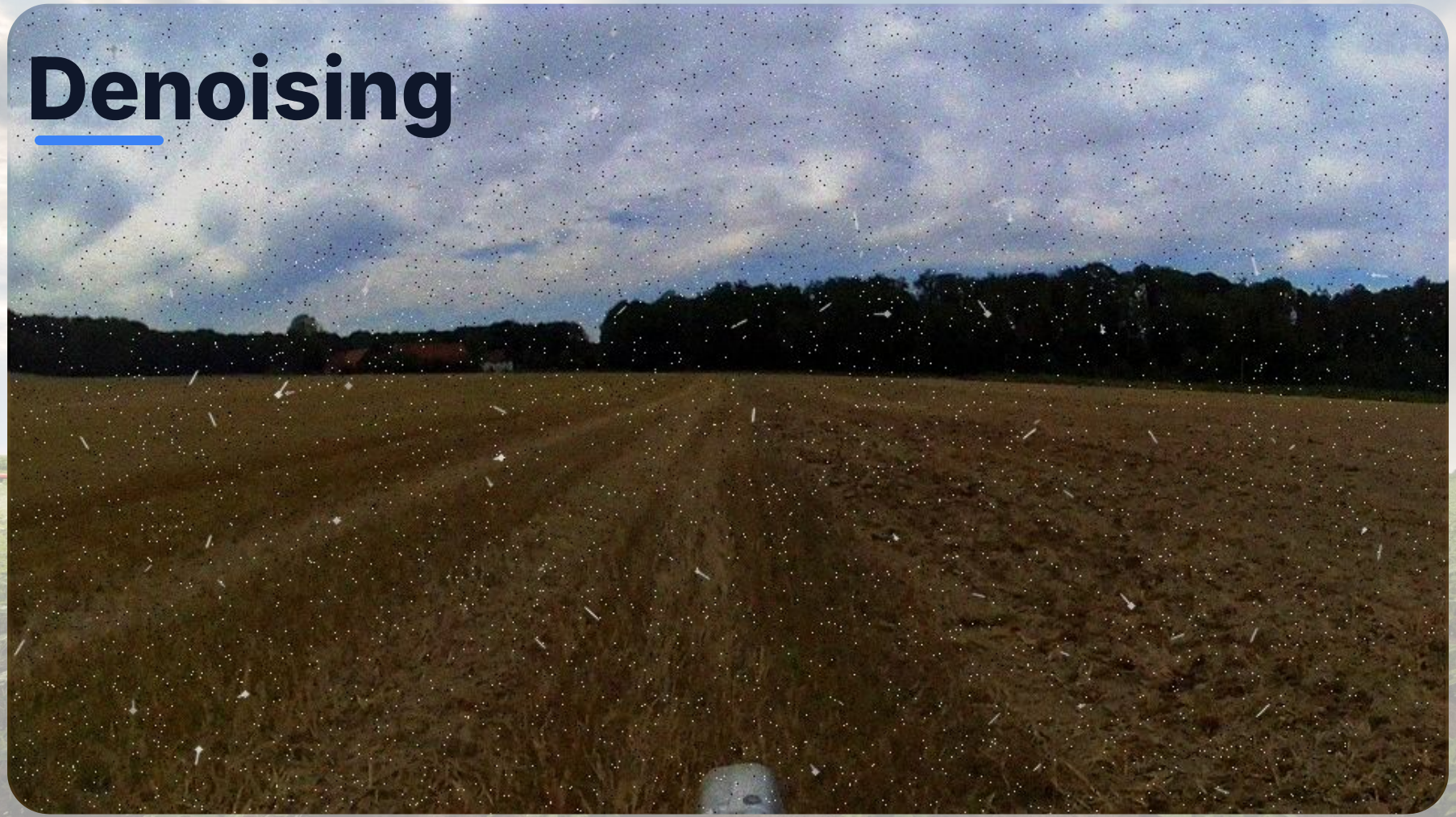
“Not scaling vertically but horizontally!”

Highlights

Denoiser
Suppresses patterns like dust or glare

Augmentation
Creating an enriched noisy dataset

Denoising



Denoising

Proof of Concept

First Layer
Input of raw images captured by the cameras

Second Layer
Real time denoising

Third Layer
Object detection by the classifier

Demonstration

Classifier Evaluation on Dirty Dataset

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 55/55 1.1it/s 51.8s
                   all        878       1782      0.546      0.389      0.401      0.169
          human_manekin       751       1782      0.546      0.389      0.401      0.169
Speed: 0.2ms preprocess, 54.0ms inference, 0.0ms loss, 0.0ms postprocess per image
```

Classifier Evaluation on Denoised Dataset

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 55/55 1.1it/s 51.0s
                   all        878       1782      0.524      0.434      0.416      0.183
          human_manekin       751       1782      0.524      0.434      0.416      0.183
Speed: 0.2ms preprocess, 54.3ms inference, 0.0ms loss, 0.0ms postprocess per image
```

Q&A Session

We are excited for your questions!